

RNAPhaseek

A foundation-model-based predictor and de novo designer
of phase-separating RNAs across cellular condensates

Comprehensive project report and paper outline

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Part 1 — From idea to result

RNAPhaseek is an end-to-end pipeline for predicting and designing RNA sequences that undergo liquid-liquid phase separation (LLPS). This report walks the project from scientific motivation through dataset construction, model training, evaluation, failed experiments, and the final design tool.

1.1 The scientific problem

Liquid-liquid phase separation of RNA is the biophysical mechanism behind membraneless cellular compartments: stress granules, paraspeckles, P-bodies, nucleoli, and neuronal RNA granules. Identifying which RNAs phase-separate matters for understanding cellular organization, neurodegeneration (TDP-43/FUS/NEAT1 in ALS), viral replication compartments, and synthetic biology applications.

Existing tools focused on protein LLPS. RNA-specific predictors were limited in scope and trained on small, single-source datasets (~390 LLPS-positive sequences in the predecessor Phaseek model). The result was modest performance (AUROC $\approx$ 0.697) and limited generalization.

The opportunity: RNA foundation models (RNA-FM, 100M params, ESM-style transformer pretrained on RNACentral) had become available, offering strong contextual representations. Combining them with a larger, multi-database training set and a lightweight task-specific head should yield a significantly better predictor, and the same predictor could be repurposed for de novo design of synthetic phase-separating RNAs.

1.2 The dataset — building the largest curated RNA-LLPS corpus to date

We assembled positives from five primary databases, each requiring different scraping, parsing, and ID-mapping strategies. Several were broken in published form and had to be fixed:

Database	Type	Yield	Status
RPS 2.0	Reviewed manually-curated LLPS RNAs	497	API response schema had changed; rewrote parser to handle columnName / columnValues array format and corrected the rpsId field.
Parker SG (GEO GSE99304)	Cuffdiff-enriched stress-granule mRNAs (U2OS)	2,282	Two-character bug fix (status == "ok" -> "OK") plus inverted fold-change convention; switched to using the classification column. Recovery via rate-limit-aware retry and dropping /xrefs/symbol/ lookups.
smOOPs (Ivanov 2025)	Mouse stress-granule transcripts -> human orthologs	1,305	xlsx parser needed to detect the real header row; Ensembl /homology/id/{species}/{id} URL had moved; MyGene /v3/querymany endpoint moved to /v3/query.
RNAPhaSep (www.rnaphasep.cn)	Curated multi-organism LLPS RNAs	217	Old domain dead; recovered new domain via DNS probing; new JSON API uses /api/show_*RNAs per RNA type; rewrote downloader to walk all 10 endpoints.
G4RNA	G-quadruplex-forming RNAs	0	Server permanently unreachable on all port/protocol combinations; Wayback only archived the landing page. Excluded.
Old-machine carryover (existing Phaseek positives)	—	959	Merged via --existing-positives flag.

Merge + CD-HIT-EST clustering at 90% identity produced 4,424 unique LLPS-positive RNAs, an order of magnitude larger than the 390-positive corpus from prior work.

Negatives were generated from Ensembl human cDNA (release-115, 221k protein-coding sequences) using reservoir sampling + GC content ($\pm 6\%$) + length ($\pm 25\%$) matching to each positive. After gene-overlap exclusion: 3,016 negatives.

1.3 The features — what the models see

For each RNA we precomputed:

- FEGS (RNA-FEGS Lhat eigenvalue matrices): top-10 per-motif ($L \times L$) walk-distance matrices encoding structural propensity. Required by base and original hybrid models as a graph-bias term in attention.
- 26-dim biophysical features (RNA2PS + ENCORI): stacking energy, condensation proxy, pairing propensity, RBP-motif counts. Fused with the pooled embedding before classification.
- RNA-FM tokenization: nucleotide-level (28-token vocab), used by hybrid models.
- BPE tokenization: character-level (~9 tokens, no BPE merges happened due to whitespace pre-tokenizer behavior), used by the base model.

1.4 The architectures — three trained models

Base model (RNAPhaseek, 5M params)

6-layer FEGSTrans transformer trained from scratch. Inputs: BPE-encoded sequences + FEGS graph bias + biophysical features. Cross-entropy with label smoothing. Output: 2-class softmax.

Hybrid Phase 1 (frozen backbone, 10M trainable)

RNA-FM backbone (frozen, 100M params, 640-dim) + 2 FEGSTrans adapter blocks operating on RNA-FM hidden states + biophysical fusion + linear head. The adapter blocks reuse the FEGS graph-bias mechanism with learnable per-block β scalars.

Hybrid Phase 2 (unfrozen backbone, 26M trainable)

Same architecture but with the last 2 RNA-FM encoder layers unfrozen. Used backbone_lr = 5e-6 for backbone params, lr = 2e-4 for adapter+head.

1.5 Training results — the headline numbers

Internal 85/15 train/val split (random_state = 42), AUROC-based early stopping with patience = 12:

Model	Trainable	Val AUROC	Val PR-AUC	β (graph-bias)	Notes
Old machine baseline	5M	0.697	0.623	—	Prior Phaseek, 390 positives
Base (this study)	5M	0.7267	0.8167	0.04 to 0.08	More data alone added +0.030
Hybrid Phase 1 (frozen)	10M	0.7707	0.8482	≈ 0	RNA-FM context added +0.044
Hybrid Phase 2 (unfreeze 2)	26M	0.7669	0.8444	≈ 0	Mildly hurt; overfits at 4,424 positives

Key biological finding: β -values converged to near zero in both hybrid runs and were large positive in the base model. RNA-FM's pretrained representations subsume the FEGS structural-bias signal; the hand-engineered structural prior is redundant when contextual embeddings are present.

1.6 The publishable test number — clean held-out evaluation

The 0.7707 val AUROC is not directly publishable because val data was used for early stopping. To produce a defensible test number we:

- Stratified split with `random_state = 999` (different from trainer's internal seed) -> 15% held-out test.
- Built subset FASTAs, FECS symlink directories, biophys arrays for the 85% train+val portion.
- Retrained Phase 1 on the 85% subset (same hyperparameters).
- Evaluated once on the untouched 15% test.

Result on 1,117 held-out RNAs (664 pos / 453 neg):

Metric	Value
AUROC	0.7642
PR-AUC	0.8463
F1 @ t = 0.20	0.7545
MCC @ t = 0.20	0.3605
Sensitivity	0.7846
Specificity	0.5673
Precision	0.7266

Comparison to baseline: +0.067 AUROC, +0.223 PR-AUC over old work; sensitivity maintained at 0.78.

1.7 The failures — what we tried that didn't work

These are scientifically informative negative results worth reporting:

Full-sequence training via multi-window aggregation

The model has a hard 1,024-token input limit (RNA-FM positional embeddings). With median training RNAs at ~4,500 nt and longest at ~33,000 nt, truncation discards information. We tested three multi-window training strategies that should, in principle, expose every nucleotide to gradient updates:

Strategy	Result	Why it failed
Max-pool over per-window embeddings	AUROC $\approx$ 0.50 across 16 epochs	Each backward pass only flows gradient through the argmax window per channel; ~30 of 32 windows get zero gradient per step; the model essentially trains on noise.
Mean-pool over per-window embeddings	AUROC stuck at 0.52 across 4 epochs	Equal weighting dilutes LLPS-driving local motifs by ~32 $\times$; discriminative signal drowns in noise from non-LLPS regions.
Attention-pool over per-window embeddings (Phase 1 init)	AUROC = 0.51 at epoch 1, declining	Phase 1's adapter+head were trained on single-window embeddings; cross-window aggregation produces input distributions the head can't predict from; LR = 5e-5 too slow to adapt.

The underlying issue: all three pooling methods produce embedding distributions fundamentally different from what RNA-FM and the trained adapter were designed for. Phase 1's "5' bias" turned out to be a feature, not a bug; LLPS-driving motifs in mRNAs cluster in 5' UTRs and gene-start regions, exactly where the truncation kept the model focused.

Phase 2 fine-tuning (unfreeze 2 backbone layers)

Predicted by the project spec to add +0.04 AUROC. Reality: AUROC slightly decreased (0.7707 -> 0.7669) while F1/MCC slightly increased. The 26M trainable parameters need more than 4,424 positives to generalize. The spec's warning ("Only do this with enough data; with few samples it overfits") was correct.

1.8 The MPS quirks — defensive engineering

Training on Apple Silicon MPS surfaced several subtle bugs requiring defensive code:

- Label corruption via host↔device round-trip. `yb.to("mps").cpu()` occasionally returned pointer-like garbage values (memory addresses in the 4-5 GB range) for ~5% of validation samples. Workaround: keep a pristine CPU copy `yb_cpu = yb.detach().clone().long()` before moving to MPS; use that for metric computation.
- Non-finite attention logits. MPS attention occasionally produces NaN/Inf for ~5% of samples (init-dependent, vanishes after a few training epochs). Workaround: sanitize logits before softmax/loss (`torch.where(non_finite, zeros, logits)`).
- OOM via FECS LRU cache. Default cache size of 4,096 items × ~40 MB = 164 GB. Reduced to 128 items (~5 GB cap).
- fp16 overflow in attention bias. $\beta \times \text{Lhat}$ exceeded fp16's 65k range as β grew during training -> softmax NaN. Disabled `fp16_bias`.
- Process group kill on session reload. Background Bash jobs died when the harness session recycled. Fixed by `os.fork() + os.setsid()` to detach training into its own session group.
- macOS Jetsam OOM kill. Multi-window training (`B=2, max_windows=32`) blew through the memory budget; reduced to `B=1, max_windows=32`.
- AMP autocast. Disabled on MPS (`amp_on = device == "cuda"`).

1.9 The de novo design tool

We adapted the SeqProp algorithm (Linder et al. 2020) from the original Phaseek to work with the hybrid model.

Mathematical core

- Initialize $\theta \in \mathbb{R}^{(L \times 4)}$ (logits per position × nucleotide).
- Forward pass: $P_{\text{soft}} = \text{softmax}(\theta / T)$; soft embedding $E = P_{\text{soft}} @ W_{\text{NT}}$ (where W_{NT} is the 4×640 RNA-FM nucleotide embedding extracted from the word-embedding table at token IDs {6, 9, 8, 7} for {A, U, G, C}), pass through `model.backbone(inputs_embeds = E)` bypassing the tokenizer entirely.
- Loss: $-P(\text{LLPS}) + \text{entropy_weight} \times \text{entropy} + \sum \text{biological-reward terms}$.
- Temperature anneals from 2.0 (exploration) to 0.1 (exploitation).
- Discrete decode via argmax, optional greedy local-search refinement.

Four differentiable biological rewards (G4 propensity, GC balance, repeat density, AU-richness) and eight condensate-specific presets (`stress_granule`, `p_body`, `nucleolus`, `nuclear_speckle`, `paraspeckle`, `germ_granule`, `cajal_body`, `neuronal_rna_granule`) with empirically-grounded reward weights.

Three generation modes: `seqprop` (classifier only), `struct` (+ biological rewards), `cond` (+ condensate preset).

1.10 The inference pipeline

predict_hybrid.py accepts a FASTA, computes FECS and biophysical features on-the-fly, scores each RNA. For long sequences (>1,022 nt), --window_mode slide --window_stride 512 --window_pool max applies a sliding-window inference: each window scored independently, per-RNA score = max across windows. Output TSV reports prob_llps, n_windows, best_window_pos (the nucleotide position where the maximum-scoring window starts; a "where's the signal" indicator).

This recovers full-sequence inference at inference time only, without the training-time pathologies that broke multi-window training.

1.11 Reproducibility artifacts

model/phase1/	<- operational Phase 1 (val AUROC 0.7707)
model/hybrid_best.pt	<- Phase 2 (val AUROC 0.7669)
model/rna_phaseek_best.pt	<- base (val AUROC 0.7267)
model/hybrid_holdout_best.pt	<- publishable Phase 1 (test AUROC 0.7642)
model/fullseq_{max,mean,attn}pool_failed/	<- negative-result archives
Data/raw/all_positives_dedup.fasta	<- 4,424 unique positives
Data/raw/negatives_ensembl.fasta	<- 3,016 GC+length-matched negatives
Data/raw/positives_heldouttest.fasta	<- 664 held-out test positives
Data/raw/negatives_heldouttest.fasta	<- 453 held-out test negatives
Data/splits/heldout_test_indices_{pos,neg}.npz	<- reproducible split (rs=999)
Functions/data_collection/	<- fixed scrapers for all 5 databases
Functions/generator_hybrid.py	<- de novo design tool (SeqProp)
predict_hybrid.py	<- inference CLI with sliding-window support

Part 2 — Paper outline

Suggested title

RNAPhaseek: A foundation-model-based predictor and de novo designer of phase-separating RNAs across cellular condensates

Suggested journal targets

- Nucleic Acids Research — natural fit (database + tool, broad audience)
- Nature Methods / Nature Computational Biology — if positioned as a methods advance (multi-database integration + foundation-model adaptation + design tool)
- Bioinformatics — if focused on the predictor only
- NAR Genomics and Bioinformatics — open-access alternative

Outline

Abstract (200–250 words)

- 2 sentences: biological motivation (LLPS importance, current gap).
- 2 sentences: prior work limitations (small datasets, modest performance).
- 3 sentences: our contribution (largest curated RNA-LLPS corpus, foundation-model adaptation, AUROC = 0.76 on held-out test, +0.07 over prior baseline, de novo design tool).
- 2 sentences: key biological findings (RNA-FM context subsumes structural priors; condensate-conditioned generation).
- 1 sentence: availability statement.

1. Introduction (~1.5 pages)

- 1.1 RNA in membraneless organelles — the biological frame.
- 1.2 The four canonical condensate classes; published examples (NEAT1/paraspeckle, TIA1/stress granule, etc.).
- 1.3 Computational landscape — protein LLPS predictors vs. the RNA gap.
- 1.4 The data scarcity problem (prior work: 390 positives) and the RNA foundation-model opportunity.
- 1.5 Our three contributions: the largest curated RNA-LLPS dataset to date (4,424 positives, 5 databases); RNAPhaseek-Hybrid, an RNA-FM-based predictor reaching AUROC 0.764 on held-out test; a SeqProp-based de novo designer for condensate-specific synthetic RNAs.

2. Materials and Methods (~3–4 pages)

2.1 Dataset construction

- Five-database aggregation strategy.
- Per-database extraction details (each as a sub-paragraph).
- CD-HIT-EST 90% identity deduplication -> 4,424 positives.
- Negative selection: Ensembl release-115 cDNA, reservoir sampling, GC ($\pm 6\%$) + length ($\pm 25\%$) matching, gene-overlap exclusion -> 3,016 negatives.
- Train/val/test stratified splits (random_state = 999); 664 + 453 = 1,117 held-out test RNAs.
- Data availability statement.

2.2 Feature extraction

- FEGS Lhat matrices (top-10 per-motif eigenvalue matrices); reference and rationale.
- 26-dim biophysical descriptors (RNA2PS + ENCORI panel); per-feature table in supplementary.
- Tokenization (RNA-FM for hybrid, BPE for base).

2.3 Model architectures

- Base RNAPhaseek: 6-layer FEGSTrans transformer, 256-dim embedding, BPE-tokenized input, FEGS attention bias with learnable β , biophysical fusion, linear head (~5M params).
- Hybrid RNAPhaseek: RNA-FM-100M backbone (frozen) + 2 FEGSTrans adapter blocks at 640-dim + bio fusion + linear head (~10M trainable).
- Architecture diagrams in supplementary.

2.4 Training

- Loss: cross-entropy with label smoothing (0.05) + L2 regularization on graph-bias mixture weights.
- Optimizer: AdamW (lr = $2e-4$ adapter, $5e-6$ backbone), cosine decay with 10% warmup.
- WeightedRandomSampler for class balance.
- AUROC-based early stopping (patience 12).
- Implementation: PyTorch + Apple MPS, batch size 4-8, ~22 min/epoch.
- Numerical-stability mitigations (defensive label handling, NaN sanitization, β fp32) reported in Supplementary Methods.

2.5 Evaluation

- Stratified held-out test set (random_state = 999, 15%, never seen during training or early stopping).
- Metrics: AUROC, PR-AUC, F1, MCC, sensitivity, specificity, precision.
- Threshold sweep on val set for F1-optimal and MCC-optimal cutoffs.
- 95% confidence intervals via DeLong's method (AUROC) or bootstrap (others).

2.6 De novo design (SeqProp)

- Mathematical formulation.
- Extraction of RNA-FM 4-row nucleotide embedding matrix.
- Differentiable biological rewards (G4, GC balance, repeat density, AU-richness).
- Condensate-specific reward presets (justify each).
- Temperature annealing schedule.
- Greedy refinement.

3. Results (~3–4 pages)

3.1 Dataset curation and statistics

- Per-source composition table.
- Length distribution (figure).
- Class balance, gene-overlap diagnostics.
- Comparison to prior RNA-LLPS datasets.

3.2 Predictor performance on held-out test

- Headline numbers: AUROC 0.7642, PR-AUC 0.8463 (Figure 2: ROC + PR curves, with baseline comparison).
- Three-way model comparison (Base / Phase 1 / Phase 2) at val.
- Threshold-dependent metrics at F1- and MCC-optimal cutoffs.

- Confusion matrix on held-out test.

3.3 Ablations and findings

- Effect of training set size (note: explicit ablation would require re-runs; we have base -> Phase 1 already showing +0.044 from RNA-FM at the same data scale).
- β -analysis: small for base (~ 0.05), near-zero for hybrid — RNA-FM context absorbs FEGS structural prior.
- Biophysical features: zero-ablation analysis (would require running with `--no_bio` flag).
- Phase 2 negative result: more capacity $\neq$ better with this data scale.

3.4 Long-sequence inference via sliding window

- Comparison: truncation vs. sliding-window aggregation on a long-RNA subset.
- Per-window score profile for known LLPS RNAs (NEAT1, MALAT1, TERRA).

3.5 De novo design demonstration

- Generated sequences from SeqProp (default), StructSeqProp, CondSeqProp.
- Score distribution histogram per method.
- GC/AU/repeat statistics: do generated sequences match natural LLPS RNA properties?
- Per-condensate generation: does `--condensate stress_granule` produce AU-richer sequences than `--condensate p_body`?
- Vienna fold-energy distributions.
- Example sequences in supplementary FASTAs.

4. Discussion (~1.5 pages)

4.1 What worked and why

- Foundation-model context dominates over hand-engineered structural priors (β -analysis evidence).
- Multi-database aggregation at $>10\times$ prior data scale enables non-trivial AUROC improvements.
- A single classifier doubles as a design tool via gradient-based optimization.

4.2 Honest limitations

- AUROC = 0.76 is meaningful but far from perfect; predictions are prioritized hypotheses, not ground truth.
- Test set drawn from same five databases as training; performance on truly novel RNA families is untested.
- Training context window of 1,024 tokens; full-sequence training did not work (Section 4.3); truly ultra-long RNAs ($>20k$ nt) handled only via sliding-window inference.
- Generated sequences are extrapolations from a model trained on natural RNAs; wet-lab validation rates may be lower than test-set AUROC suggests.

4.3 The full-sequence training negative result

- Three pooling strategies (max, mean, attention) all failed.
- Underlying issue: cross-window aggregation produces input distributions Phase 1's head can't predict from.
- Sliding-window inference recovers long-RNA prediction at inference time without the training pathology.
- Implication: future work should explore longer-context RNA foundation models, not training-time multi-window aggregation.

4.4 The Phase 2 negative result

- More parameters ≠ better with this data scale.
- Suggests future improvement requires more positives, not more capacity.

4.5 Outlook

- Hard-negative mining (negatives that look superficially like positives).
- Integration with longer-context RNA foundation models (Nucleotide Transformer, etc.).
- Condensate-resolved training labels (per-condensate classifiers) once data permits.
- Wet-lab validation of designed sequences.

5. Conclusion (~half page)

- Largest curated RNA-LLPS dataset to date.
- State-of-the-art predictor with rigorous held-out evaluation.
- Practical de novo design tool with condensate-specific conditioning.
- Pretrained RNA representations subsume hand-engineered structural priors — guidance for future biological deep-learning architectures.

Supplementary materials

- S1: Per-database extraction protocols and the fixes required.
- S2: Numerical-stability strategies on Apple Silicon MPS.
- S3: Full training logs (per-epoch metrics) for all four runs.
- S4: Architecture diagrams (Base, Phase 1, Phase 2, generator soft-forward path).
- S5: Per-feature table for the 26-dim biophysical panel.
- S6: 8 condensate presets with biological justification per preset.
- S7: Full-sequence training experiments (the three failed pooling strategies).
- S8: Threshold sweep curves; sensitivity-specificity trade-offs.
- S9: Selected generated sequences (FASTA) with predicted properties.
- S10: Reproducibility — split indices, hyperparameters, software versions.

Code & data availability

- GitHub repository (the entire RNAPhaseek_scripts/).
- Trained model checkpoints on Zenodo or HuggingFace.
- Test set indices in the repo (reproducible from random_state = 999).
- Inference and design CLIs documented in README.

Acknowledgments and funding

- Apple silicon compute donation (if relevant).
- Authors of the five source databases.
- Authors of RNA-FM (multimolecule, Zhang et al.) and ViennaRNA.
- INSERM funding.

Suggested figures (5–6 main)

- Figure 1: Pipeline overview — schematic of dataset → training → evaluation → inference + design.
- Figure 2: Dataset composition — Venn diagram of source overlap + length distribution + class balance.

- Figure 3: Model performance — ROC curve + PR curve on held-out test, baseline comparison; per-model bar chart.
- Figure 4: β -value analysis — learned graph-bias weights across base / Phase 1 / Phase 2, with interpretation.
- Figure 5: Long-RNA sliding-window inference — score profile across NEAT1 (or similar) showing where the model finds LLPS signal.
- Figure 6: De novo design — generated sequence properties (GC, AU, repeat density, fold energy) by condensate preset.